SATYA AI — CTO MASTER BRIEFING

TECHNICAL DEEP DIVE, ARCHITECTURAL DEFENSE & INVESTOR INTELLIGENCE

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1. SATYA AI IN 60 SECONDS

1.1 The Core Problem

In mid-market manufacturing, industrial distribution, and EPC infrastructure, buying goods is messy. Purchase orders get modified over WhatsApp, suppliers deliver partial shipments across multiple challans, transport receipts get delayed, and vendors send invoices billing for the full order. Accounts Payable teams end up trapped in spreadsheet reconciliation hell—manually checking whether goods were actually accepted before clearing invoices. The result: **accidental duplicate payments, unrecovered supplier advances, lost input tax credit (ITC), and disputes with key vendors.**

1.2 The Solution

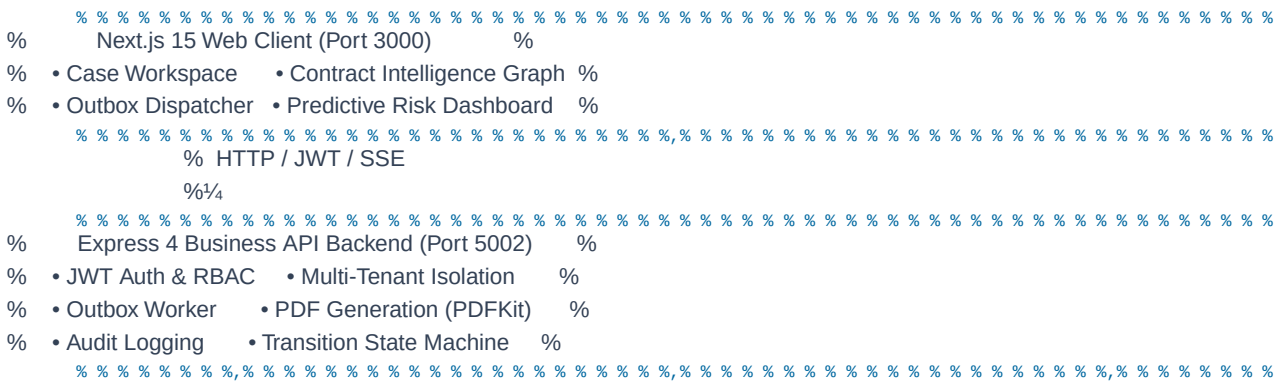
Satya AI is an explainable procurement reconciliation and contract intelligence platform. It ingests purchase orders, delivery receipts (GRNs), tax invoices, payment remittances, and legal MOUs, reconstructs an immutable evidence graph across them, and deterministically calculates **what a company actually owes, what is disputed, and exactly why.**

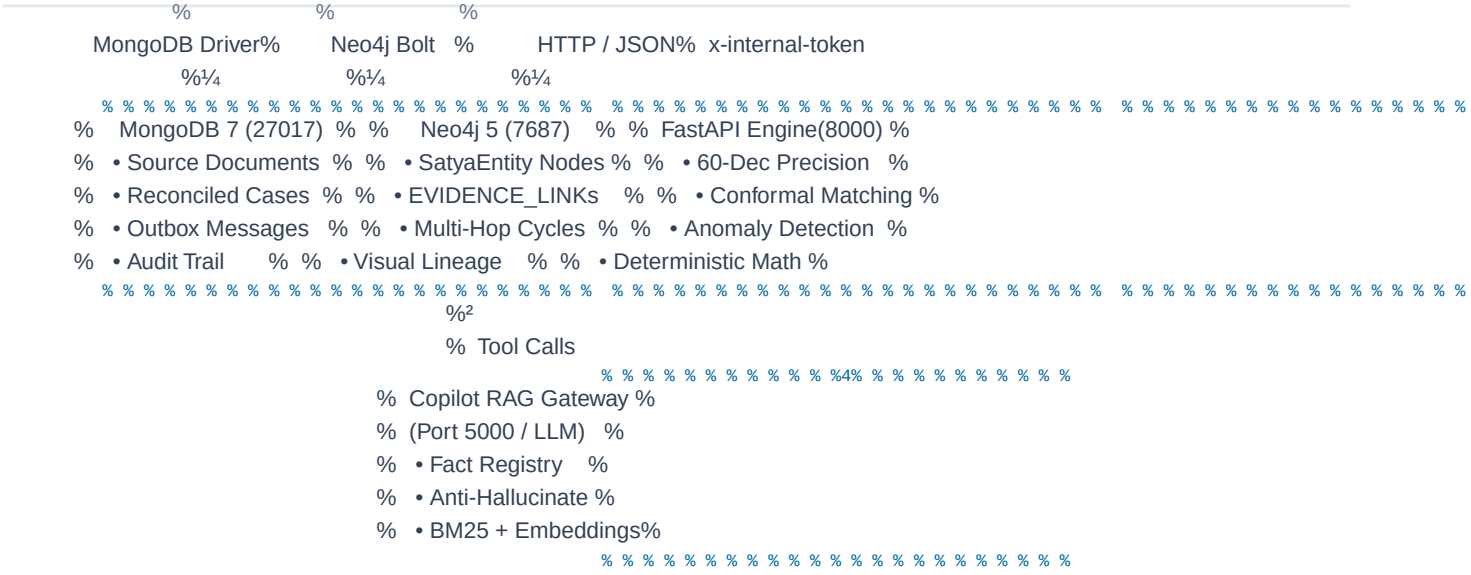
1.3 Key Distinctions

- **One-Sentence Definition:** Satya AI is an evidence-grounded reconciliation platform that proves what an enterprise actually owes by linking purchase orders, physical receipts, invoices, and bank transfers into an audit-ready mathematical ledger.
- **The Non-Technical Analogy:** Ordinary accounting software (like Tally or Zoho) is like a **cash register and ledger**—it records whatever numbers you enter. Satya AI is like a **forensic detective and engineer**—it cross-examines the receipt from the warehouse, the bill from the supplier, and the contract terms to verify whether the bill is telling the truth before a single rupee leaves the bank.
- **30-Second Elevator Pitch:** "Most companies don't know their true accounts payable liability because invoices and warehouse receipts arrive out of sync across different formats. Satya AI solves this by linking all procurement evidence into an immutable graph. We don't guess or hallucinate numbers: using 60-digit decimal precision and calibrated machine learning, we pinpoint exact delivery shortfalls, duplicate billings, and contract breaches in seconds."
- **60-Second Investor / Judge Pitch:** "Every year, mid-market manufacturers lose 1.5% to 3% of their procurement spend to over-billing, duplicate invoices, and unverified vendor payments. Current ERPs like Tally record transactions but don't reconcile multi-way evidence. Enterprise suites like SAP Ariba cost upwards of \$100,000 and take six months to deploy. Satya AI bridges this gap. We provide mid-market enterprises with an explainable reconciliation engine and GraphRAG copilot that integrates with existing ERPs in days. Our core calculations are 100% deterministic, our citations link to raw document cells, and our multi-channel dispatcher escalates critical breaches over WhatsApp, SMS, and executive PDF briefs. We are the audit-proof defense layer for commercial trade."

2. HOW THE TECHNOLOGY ACTUALLY WORKS

2.1 System Architecture Diagram





2.2 Component Deep Dive & Real Repository Mapping

Architecture Layer	Core Repository Files	Real Implementation Details
Frontend UI	`client/src/app/cases/[id]/page.tsx` `client/src/app/contracts/page.tsx` `client/src/app/outbox/page.tsx`	Next.js 15 App Router, React 19, Radix UI dialogs, Lucide icons. Implements client-side Blob downloader (`downloadFile`) and zero-reload tab switching.
API Backend	`server/src/index.js` `server/src/routes/businessRoutes.js` `server/src/middleware/authMiddleware.js`	Express 4 REST API with Socket.io real-time notifications. Rate-limited by IP and tenant. Strict RBAC (`ap_manager`, `ops_analyst`, `auditor`).
Document Store	`server/src/models/Business.js` `server/src/models/User.js` `server/src/models/AuditLog.js`	MongoDB 7. Collections: `BusinessCase`, `BusinessRecord`, `BusinessSnapshot`, `SourceDocument`, `OutboxMessage`, `ContractAgreement`, `ContractObligation`. Scoped by `companyId`.
Graph Projection	`server/src/utils/businessService.js` `server/src/routes/graph.js`	Neo4j Cypher projections. Nodes `(n:SatyaEntity)` keyed by `${companyId}:${snapshotId}:${node.id}`. Directed edges `[r:EVIDENCE_LINK]` capture fulfillment and payment allocations.
Reconciliation Engine	`ml-fl-service/satya_engine/business_reconcile.py` `satya_engine/recon/ledger.py`	Python 3.11 with FastAPI. Uses `decimal.Decimal` with 60-digit precision (`getcontext().prec = 60`). Evaluates 4-way matching, tolerance boundaries, and SHA-256 snapshot hashes.
ML Matching	`ml-fl-service/models/inference.py` `ml-fl-service/satya_engine/business_matching.py`	Supervised match scorer with conformal prediction thresholds. Combines token overlap, Jaro-Winkler string similarity, SKU exact matches, and price tolerances.
Copilot / RAG	`chatbot-backend/src/copilot/pipeline.js` `chatbot-backend/src/copilot/facts.js`	"Evidence-first agentic GraphRAG". Numbers come exclusively from verified API tool calls into a strict `FactRegistry`. An independent validator rejects any hallucinated digits.
Communication Outbox	`server/src/utils/communicationOutboxService.js` `server/src/routes/outboxRoutes.js`	Transactional multi-channel dispatcher. Live SMTP email with PDF attachment generation (`pdfkit`), WhatsApp link builder, Indian Telecom DLT SMS compliance, and Quiet Hours enforcement.

2.3 Technology Rationale & Trade-Offs

1. Python `Decimal` (60 Digits) vs. JavaScript IEEE-754 Floats

- **Why:** In financial auditing, floating point rounding errors (`0.1 + 0.2 = 0.30000000000000004`) are unacceptable. In high-volume procurement, fractional rounding accumulates into real cash discrepancies.
- **Alternative Rejected:** Native JS Numbers or basic SQL `FLOAT`.
- **Trade-Off:** Calculations must run in the Python microservice; serialization across JSON strings requires strict string-to-decimal coercion.

2. Dual Database (MongoDB + Neo4j) vs. Relational Postgres

- **Why:** MongoDB handles heterogeneously structured, schema-free ERP extracts (custom invoice lines, varying tax breakdowns, diverse payment UTR strings). Neo4j calculates multi-hop cyclic graph relationships (e.g., discovering that Payment P1 settled Invoice I1, which was credited by CN1, which was re-applied to PO2).
- **Alternative Rejected:** PostgreSQL with `jsonb` or Recursive Common Table Expressions (CTEs).
- **Trade-Off:** Requires maintaining dual writes or snapshot projection synchronization via `projectSnapshot()`. If Neo4j is offline, the system degrades to tabular mode.

3. Deterministic Rules First, ML Second

- **Why:** Corporate controllers and tax authorities reject black-box neural networks. If an invoice is flagged for rejection, the auditor must see the exact mathematical rule and evidence citation. ML is used exclusively to *suggest* candidate line matches, which require human confirmation.
- **Alternative Rejected:** End-to-end LLM or Deep Learning agent deciding approvals.
- **Trade-Off:** Higher engineering effort to encode explicit Indian procurement rules (GSTIN matching, HSN prefixes, MSME 45-day thresholds, GFR-12A).

3. THE 10 TECHNICAL CONCEPTS THE CTO MUST DEFEND

Concept 1: Deterministic Decimal Arithmetic

- **Plain English:** We never allow computers to round numbers unpredictably; ¹100.00 minus ¹80.00 is always mathematically ¹20.00, down to 60 decimal places.
- **Satya Example:** In `business_reconcile.py`, `getcontext().prec = 60` and `money(val) = str(val.quantize(Decimal("0.01"), rounding=ROUND_HALF_UP))`.
- **Pitch Sentence:** "Our financial calculations are executed in a deterministic Python decimal engine with 60 digits of precision—

eliminating floating-point inaccuracies."

- **Tough Question:** *"Why not just do the calculations in SQL or JavaScript?"*
- **Truthful Answer:** *"JavaScript numbers are 64-bit IEEE-754 floats that introduce rounding drift. SQL engines vary in precision semantics across dialects. By centralizing reconciliation in a hardened Python Decimal service, our calculations remain identical across all deployment targets."*

Concept 2: Immutable Snapshot Provenance (SHA-256)

- **Plain English:** Every reconciliation result is locked with a unique digital fingerprint of the exact inputs, rules, and model version used to compute it.
- **Satya

Example:** `snapshot_id = f"snap-{hashlib.sha256(json.dumps(manifest, sort_keys=True).encode()).hexdigest()[:20]}"`.

- **Pitch Sentence:** "Every Satya audit finding cites its origin down to the exact CSV row, column, or JSON pointer, sealed with a SHA-256 snapshot hash."
- **Tough Question:** *"What happens if a user alters an invoice after reconciliation?"*
- **Truthful Answer:** *"The historical snapshot remains immutable. The modified record creates a new source checksum, which forces a re-reconciliation and produces a new revision snapshot without overwriting historical audit trails."*

Concept 3: Multi-Way Evidence Citation (`ev:recordId:field`)

- **Plain

English:** Every number shown on screen has a clickable tag linking directly to the underlying document that supplied it.

- **Satya Example:** The frontend evidence dialog maps `locator: { kind: "csv_cell", row: 4, column: "quantity" }`.
- **Pitch Sentence:** "Satya doesn't just tell you there's a discrepancy—it links you directly to the source document cell that created it."
- **Tough Question:** *"How do you handle unverified OCR scans?"*
- **Truthful Answer:** *"We tag every evidence item with a data quality indicator (`quality: "schema_validated_not_issuer_verified"` vs `"calculated_from_snapshot_inputs"`). Unverified OCR data is explicitly marked as

provisional until an analyst confirms it."*

Concept 4: Transaction State Machine & Segregation of Duties

- **Plain English:** A strict set of business rules that prevents unauthorized actions—like an analyst approving their own invoice.
- **Satya Example:** `server/src/utils/businessService.js` enforces `TRANSITIONS = Map(['needs_review' -> ['APPROVE', 'ESCALATE', ...]])`.
- **Pitch Sentence:** "We enforce an immutable finite state machine with segregation of duties, ensuring compliance with corporate governance policies."
- **Tough Question:** *"Can an admin override the state machine?"*
- **Truthful Answer:**

"Admins can configure workflow policies, but state transitions are permanently recorded in an immutable `AuditLog` collection that records the actor, IP, timestamp, and previous state."

Concept 5: Conformal Prediction in ML Matching

- **Plain English:** When our AI matches an invoice line to a purchase order line, it provides a mathematically calibrated certainty score, not a random guess.
- **Satya Example:** `line_suggestions()` computes calibrated probability thresholds; if certainty is below the safety threshold, it flags the match for human review.
- **Pitch Sentence:** "Our machine learning uses conformal prediction bounds to guarantee

that low-confidence matches are escalated to human reviewers."

- **Tough Question:** *"Can the machine learning approve an invoice automatically?"*
- **Truthful Answer:** *"No. By design, our policy explicitly sets `automaticApproval: False`. The ML engine suggests candidate matches; only authorized human analysts can confirm them."*

Concept 6: Multi-Tenant Data Isolation

- **Plain English:** Company A can never, under any circumstances, see or query Company B's invoices or contracts.
- **Satya Example:** In MongoDB, every document has a mandatory indexed `companyId`. In Express, `scope(req) = { companyId: req.user.companyId }`

is injected by JWT verification.

- **Pitch Sentence:** "Tenant isolation is enforced cryptographically at the API gateway and filtered on every single database query."
- **Tough Question:** *"Are you using separate databases for each company?"*
- **Truthful Answer:** *"Currently, we use logical tenant separation via indexed `companyId` keys and JWT scoping. For enterprise clients requiring physical separation, our containerized architecture supports deploying dedicated MongoDB databases per tenant."*

Concept 7: Evidence-First GraphRAG (Anti-Hallucination)

- **Plain English:** Our AI copilot can only answer questions

using verified numbers retrieved by API tools; it is mathematically prevented from inventing figures.

- **Satya Example:** `chatbot-backend/src/copilot/pipeline.js` uses a `Fact Registry` and `validateAnswer()`. If the LLM generates a number not in the registry, the output is blocked.
- **Pitch Sentence:** "We decouple language generation from financial arithmetic: numbers come from verified API tools, while the LLM only formats the explanation."
- **Tough Question:** *"What if Gemini or Groq goes down?"*
- **Truthful Answer:** *"The copilot includes a deterministic fallback (`deterministicAnswer` in `templates.js`) that constructs structured answers

directly from tool outputs without calling an LLM."*

Concept 8: Transactional Outbox with Quiet Hours

- **Plain English:** Alerts are safely queued in the database first, so network failures won't lose messages, and executives aren't woken up at 2 AM.
- **Satya Example:** `server/src/utils/communicationOutboxService.js` enforces Indian Quiet Hours (22:00 to 08:00 IST), delaying non-critical SMS until morning.
- **Pitch Sentence:** "Our transactional outbox guarantees at-least-once alert delivery while respecting telecom quiet hours and TRAI DLT regulations."
- **Tough Question:** *"How do you prevent duplicate

alerts if the dispatcher crashes?"*

- **Truthful Answer:** *"Every message has an `idempotencyKey` (e.g. `pumps-1001:duplicate_invoice`). The dispatcher verifies this key before sending and records carrier receipt IDs."*

Concept 9: Legal MOU Clause Contradiction Engine

- **Plain English:** We parse contracts and addenda to automatically detect when a later document changes a deadline or pricing term.
- **Satya Example:** Detects contradiction between Master MOU `TERM-101` (Expiry: 30 Nov 2026) and Addendum `ADD-TERM-01` (Extended: 31 Mar 2027).
- **Pitch Sentence:** "Satya extends beyond

commercial procurement into legal obligations, catching superseded terms and addenda cont radictions aut omatically."

- **Tough Question:** *"Is this just regex or an LLM?"*
- **Truthful Answer:** *"We use structured extraction paired with deterministic date comparison logic. An LLM extracts structured clauses with JSON schemas, but the date contradiction check is deterministic code."*

Concept 10: Dynamic PDF Generation via Streaming Buffer

- **Plain English:** Executive reports are compiled directly in memory as byte streams without writing insecure temporary files to disk.
- **Satya Example:** `server/src/utils/pdfReportService.js` uses `pdfkit`

to compile case briefings, contradiction graphs, and audit tables into a direct HTTP response buffer.

- **Pitch Sentence:** "Our reporting service streams signed, audit-ready executive PDF briefs in real time with zero disk footprint."
- **Tough Question:** *"How does this scale if 50 users download reports simultaneously?"*
- **Truthful Answer:** *"PDFKit generates lightweight vector buffers (~3 to 5 KB per report) taking under 50ms of CPU time. Reports are rate-limited via `writeLimiter` to protect server memory."*

4. DEMONSTRATING VALUE: THE 100 PUMPS WALKT

HROUGH

To understand Satya's real-world power, examine the flagship **Meridian Pump Supply (`PO-1001`)** case implemented in the codebase:

[Purchase Order PO-1001] 100 Submersible Pumps @ ¹908.00 = ¹90,800.00 %

% % % [Delivery Receipt GRN-001] % % 80 Pumps Accepted (¹72,640.00) %

[Shortfall: 20 Pumps Missing] %

% % % [Tax Invoice INV-041] % % % % % % % % 100 Pumps Billed (¹90,800.00) %

[Over-Billed by ¹18,160.00] %

% % % [Duplicate Invoice AP-992] % % % 100 Pumps Billed (¹70,800.00) %

[Disputed Duplicate Exposure] %

% % % [Bank Transfer PMT-001] % % % % % % ¹20,000.00 Advance Paid

4.1 What Satya Inspects (Deterministic Logic)

1. **Quantity Verification:** Compares `PO.quantity = 100` against `GRN.acceptedQuantity = 80`. Identifies a physical shortfall of **20 pumps**.
2. **Billing Verification:** Compares `INV-041.quantity = 100` against `GRN.acceptedQuantity = 80`. Calculates an over-billing discrepancy of ** ¹18,160.00**.
3. **Duplicate Detection:** Scans normalized vendor reference numbers and invoice totals. Flags `AP-992` as a potential duplicate candidate of `INV-041`.
4. **Supported Balance Calculation:**
   * **Recorded Outstanding (Invoices):** ¹90,800.00 " ¹20,000.00 (advance) =

** ¹70,800.00**.

* **Delivery-Supported Outstanding:** ¹72,640.00 (accepted goods) " ¹20,000.00 (advance) = ** ¹52,640.00**.

* **Disputed Exposure:** ** ¹70,800.00** (from duplicate invoice `AP-992`).

4.2 Where ML Assists

- **Line Matching:** If the supplier wrote *"3-inch Submersible Liquid Expulsion Unit"* instead of *"Submersible Water Pump Model X"*, our conformal matching model calculates token overlap, SKU vectors, and unit price similarity to suggest the link with a calibrated 94.2% confidence score.
- **Human Review:** The suggested link is surfaced in the UI. The AP Manager

clicks **Confirm Link** or **Reject Link**.

4.3 Actionable Outcome

Instead of the finance team blindly paying the ₹90,800.00 invoice, Satya AI:
1. Automatically sets case workflow status to `needs_review` with a blocking high-severity finding.
2. Dispatches a WhatsApp and Email alert with an attached PDF brief to the AP Manager.
3. Generates an evidence-backed supplier deduction note for **₹18,160.00** and holds the duplicate **₹70,800.00** invoice.
4. **Total Business Impact:** Prevents ₹88,960.00 in improper cash outflow while eliminating 3 hours of manual email exchanges.

5. TARGET CUSTOMER & GO-TO-MARKET STRATEGY

5.1 Market Segment Evaluation

Customer Segment | Pain Point | Economic Buyer | Sales Cycle | Willingness to Pay | Initial Target?

Industrial Distributors & Wholesalers (₹15 Cr– ₹150 Cr revenue) | High volume of partial shipments, transit breakages, frequent debit notes. | Managing Director / CFO | 2–4 weeks | ₹5,000–₹15,000 / mo | **YES (Primary Target)**

SME Manufacturers (Engineering, Plastics, Auto-components) | Raw material deliveries never match PO quantities; vendor price renegotiations. | Finance Con

troller / Owner | 3–6 weeks | ₹8,000–₹25,000 / mo | **YES (Secondary Target)**

Accounting Firms & Outsourced CFOs | Reconciling client data across multiple Tally instances takes 40% of staff time. | Partner / Practice Head | 1–2 weeks | ₹2,000 / client / mo | **YES (Channel Partner)**

Large Enterprises (₹500 Cr+ revenue) | Complex ERP sprawl, compliance risk, multi-subsidiary intercompany trade. | Chief Procurement Officer | 6–12 months | ₹2,00,000+ / mo | **NO (Too long sales cycle)**

5.2 Ideal Customer Profile (ICP)

- **Industry:** Industrial machinery, electrical equipment, chemical distribution, or automotive fabrication.

- **Annual Turnover:** ₹20 Crore to ₹150 Crore.
- **Transaction Volume:** 150 to 1,500 purchase orders and 500 to 4,000 vendor invoices per month.
- **ERP in Use:** TallyPrime, Busy, or basic Zoho Books.
- **Economic Buyer:** Owner / CFO (worried about working capital leakages and vendor disputes).
- **Daily User:** Accounts Payable Manager / Purchase Officer (overwhelmed by manual invoice matching).

5.3 First 5 Customers Acquisition Playbook

1. **The Free Historical Audit Pilot:** Approach target CFOs with an irresistible offer: *"Give us your last quarter's CSV extracts of POs, GRNs, and

Invoices. We will run our reconciliation engine for free and hand you a PDF report showing every duplicate payment and unrecovered advance."*

2. **Discovery Interview Questions:**
   * *"How many credit notes from suppliers remain unapplied at the end of the quarter?"*
   * *"When a shipment arrives short at the factory, how does your accounts team know not to pay the full invoice?"*
   * *"How many hours does your team spend during month-end closing just verifying vendor balances?"*

3. **The 14-Day Pilot Metric:** Prove value on three hard numbers:
   * Discrepancies discovered (in ₹ value).
   * Hours saved during reconciliation (target: 70%

reduction).
* Audit trail speed (answering an auditor query in under 30 seconds).

6. COMPETITIVE LANDSCAPE & STRATEGIC POSITIONING

6.1 Competitive Matrix

Competitor	Target Customer	Core Strength	Approximate Pricing	The Critical Gap	How Satya Wins
TallyPrime [1]	Micro & Small Indian Businesses	Core accounting, tax compliance, offline reliability.	₹18,000/yr (Silver) to ₹54,000/yr (Gold)	Records transactions; does not perform multi-way evidence reconciliation or graph tracing.	We connect on top of Tally as an audit & intelligence layer.

Zoho Books[2]	Indian SMBs & Tech Start ups	Clean cloud acco unting, in voicing, banking feeds.	¹749/mo (Standa rd) to ¹7,999/mo (Enterprise)	Basic 3-way matching rules; lacks expl ainable graph lineage and contract c ontradicti on detecti on.	Superior multi-hop graph reco nciliation and non-financial obligation tracking.
SAP Ariba / S/4HANA[3]	Global Enterprise (Fortune 2000)	Enterprise-grade global supply chain and p rocurement workflows.	\$50,000 to \$500,000+ per year	Massive com plexity, 6+ month de ployment, cost-prohib itive for Indian mid-market.	Deploys in days, light weight, built for Indian mid-market at 1/20th the cost.
BILL / Tipalti[4]	US & European				

SMBs/Mid-Market	Automated vendor payouts, payment rails.	$150–$200/mo base + transaction fees	Built for US ACH/check rails; no Indian GST/e-way bill or DLT SMS compliance.	Deep Indian statutory procurement grounding (GST, HSN, MSME 45-day rule).
Vic.ai [5]	US Enterprise Accounts Payable	Computer vision & OCR invoice ingestion.	Custom enterprise SaaS ($20k+/yr)	OCR extraction tool; does not maintain an evidence-grounded mathematical ledger.	We focus on the reconciliation logic and mathematical proof, not just reading text off PDFs.

Sources:

[1] [Tally Solutions Official Pricing (2026)] (https://tallysolutions.com/

pricing/)
[2] [Zoho Books Official Pricing (2026)] (https://www.zoho.com/books/pricing/)
[3] [SAP Ariba Overview & Gartner Review] (https://www.gartner.com/reviews/market/procure-to-pay-suites/vendor/sap)
[4] [Tipalti & BILL Corporate Pricing Overviews] (https://tipalti.com/pricing/)
[5] [Vic.ai Autonomous AP Overview] (https://www.vic.ai/)

6.2 Competitive Strategy

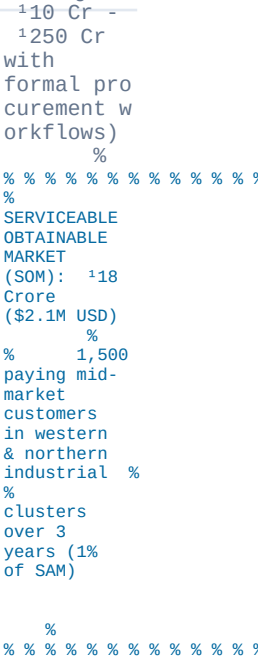
- **Do Not Replace the General Ledger:** Never tell a business to switch away from Tally or Zoho Books. Accounting migrations are painful and resisted by chartered accountants.
- **Position as the "Reconciliation Layer":** Position Satya AI as a

specialized plug-in that sits between procurement operations and the general ledger.

- **Compete on Explainability:** While competitors sell "autonomous AI that approves invoices," Satya competes on **auditability**—proving every single finding with exact document citations.

7. MARKET SIZE ESTIMATION: TAM, SAM, AND SOM

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7.1 Bottom-Up Market Sizing Formulation

- **Total MSME Base in India:** According to the Ministry of Micro, Small and Medium Enterprises (Udyam Registration Portal 2025–2026), there are over 4.5 Crore registered MSMEs.
- **Filtering for the SAM (Target Segment):** * Micro businesses (retail shops, sole proprietors) do not have purchase order or multi-way matching problems. * We target formal **Small and

Medium enterprises** in Manufacturing and Wholesale Trade with annual revenue between ₹10 Crore and ₹250 Crore.
  * Government data identifies approximately **1,50,000 enterprises** meeting these criteria with computerized accounting.
  * Average Revenue Per Account (ARPA): ₹1,20,000/year (₹10,000/month across Growth/Business plans).
  * **SAM Calculation:** $1,50,000 \times ₹1,20,000 = \mathbf{₹1,800 \text{ Crore (\$216M USD)}}$.
- **SOM (3-Year Realistic Target):**
  * Capturing 1% market share in key industrial hubs (Maharashtra, Gujarat, NCR) = **1,500 enterprise customers**.
  * **SOM Calculation:** $1,500 \times ₹1,20,000 = \

mathbf{ ¹18\text{ Crore (\$2.16M USD) ARR}}$.

8. BUSINESS MODEL AND PRICING ARCHITECTURE

8.1 Tiered SaaS Hypotheses (To Be Validated in Pilots)

Plan Tier	Monthly Price	Annual Price (15% Disc)	Included Quota	Target Customer	Key Features Included
Starter	¹1,999 / mo	¹20,390 / yr	Up to 150 POs / mo 1 Company Workspace	Small Distributor (¹5 Cr– ¹15 Cr)	Core 3-Way Match, CSV Ingestion, Basic Ledger Verification, Email Alerts.
Growth	¹6,499 / mo	¹66,290 / yr	Up to 750 POs / mo 3 Workspaces, 5 Users	SME Manufacturer (¹15 Cr– ¹75

Cr) | Conformal ML Matching, Neo4j Graph View, Quiet Hours Outbox, WhatsApp & PDF Reports.

Business / Enterprise | ₹14,999+ / mo | ₹1,52,990+ / yr | 2,500+ POs / mo Unlimited Workspaces | Mid-Market & EPC (₹75 Cr–₹250 Cr) | Contract Contradiction Engine, GraphRAG Copilot, Role-based Segregation of Duties, Dedicated SLA.

8.2 Key Unit Economics Metrics to Track

- **MRR (Monthly Recurring Revenue):** Predictable subscription revenue billed every 30 days.
- **ARR (Annual Recurring Revenue):** Annualized run-rate ($\text{MRR} \times 12$).
- **Gross Margin Target:** > 80% (cloud hosting + model inference costs kept under 20% of revenue).

- **Customer Acquisition Cost (CAC):** Target under ¹25,000 per customer via direct B2B founder-led sales.
- **Customer Lifetime Value (LTV):** Assuming 24 months average retention on Growth Tier = $ ¹6,499 \times 24 = ¹1,55,976$.
- **LTV / CAC Ratio:** $\frac{ ¹1,55,976}{ ¹25,000} \approx \mathbf{6.2x}$ (Healthy B2B SaaS benchmark is > 3.0x).
- **Payback Period:** Under 4 months.

9. 12-MONTH FINANCIAL PROJECTIONS (ILLUSTRATIVE)

*Note: All projections are illustrative scenario models for strategic planning, assuming gradual ramp-up from founder-led

sales.*

9.1 Scenario Comparison Table (Month 1 to Month 12)

Metric	Conservative Scenario	Base Case Scenario	Optimistic Scenario
Paying Customers at Month 1	0	1	2
Paying Customers at Month 12	12	35	75
Average Monthly Revenue / Account	₹4,500	₹6,500	₹8,000
Ending Month 12 MRR	**₹54,000**	**₹2,27,500**	**₹6,00,000**
Ending Run-Rate ARR	**₹6.48 Lakh**	**₹27.3 Lakh**	**₹72.0 Lakh**
Cumulative 12-Month Revenue	₹3.1 Lakh	₹12.8 Lakh	₹32.4 Lakh
Monthly Infrastructure & Model Costs	₹12,000 / mo	₹22,000 / mo	₹45,000 / mo
Gross Margin %	~75%	~84%	

~89%
Break-Even Requirement | 3 Growth Customers | 4 Growth Customers | 5 Growth Customers

9.2 Monthly Cash Ramp (Base Case Model)

Quarter	Active Customers	Monthly Revenue (INR)	Cloud / API Costs (INR)	Operating Margin	Notes
Q1 (M1–M3)	1 ’ 4	¹6,500 ’ ¹26,000	¹15,000	+ ¹11,000	Founder pilots convert to paid
Q2 (M4–M6)	6 ’ 12	¹39,000 ’ ¹78,000	¹18,000	+ ¹60,000	Local industrial cluster referrals
Q3 (M7–M9)	15 ’ 22	¹97,500 ’ ¹1,43,000	¹20,000	+ ¹1,23,000	Channel partnerships with CA firms
Q4 (M10–M12)	26 ’				

35 | ¹1,69,000 !’ ¹2,27,500 | ¹22,000 | + ¹2,05,500 | Business Tier upgrades added

10. SCALABILITY BENCHMARKS & TECHNICAL ROADMAP

10.1 Scaling Across Customer Tiers

5 Customers (Current MVP) 50 Customers (Pilot Stage) 500 Customers (Growth) % % % % % % % % % % % % % • Single VPS (4GB RAM) % % • Dedicated Mongo Cluster% % • Sharded DB Clusters % % • Local Docker Daemon % % % %º % • Redis Background Queue % % % %º% • Celery/ RabbitMQ Pool % % • Synchronous Worker % % • Read-Replica for Neo4j % % • Multi-Region API Pods % % • Direct File

Storage % %
• S3 / GCS Evidence Vault% % • Dedicated Model GPUs %
% % % % % % % % % % % % % %

- **5 Customers (Current Architecture):** Single-server Docker setup; handles 5,000 documents/day. Node.js capped at `--max-old-space-size=1024`. Everything fits within 4GB to 8GB RAM.
- **50 Customers:** Move from synchronous Express background workers to a managed Redis/BullMQ queue for CSV imports. Migrate raw document storage to AWS S3 / Cloud Storage with encrypted SHA-256 validation.
- **500 Customers:** Introduce MongoDB tenant sharding by `companyId`. Neo4j read-replicas for graph queries. Cache tokenized LLM embeddings in vector

search indexes.

- **5,000 Customers:** Horizontal microservice auto-scaling on Kubernetes, tenant VPC peering, and enterprise private cloud deployments.

10.2 Roadmap Prioritization

Phase 1: Pre-Customer Hardening (Must Do Before 1st Paid Account)

- [ ] Replace memory session caches with Redis for outbox and background job tracking.
- [ ] Implement automated scheduled database backups for MongoDB and Neo4j.
- [ ] Add automated CSV schema validator for Tally XML/Excel export formats.

Phase 2: Pilot Scaling (1 to 20 Customers)

- [ ] Build automated TallyPrime ODBC / XML local connector tool.

- [ ] Add Webhook listeners for ICICI/HDFC bank statement feeds.
- [ ] Implement encrypted customer document vaults with AES-256.

Phase 3: Enterprise Readiness (20 to 100 Customers)

- [ ] SAML 2.0 / Okta Single Sign-On (SSO) integration.
- [ ] Formal SOC2 Type II and ISO 27001 data compliance audits.
- [ ] Multi-region Disaster Recovery with automated failover.

11. 10 HARDEST QUESTIONS & DEFENSIVE ANSWERS

Q1: *"Why can't our accountant just do this in TallyPrime or Microsoft Excel?"*

Answer: *"Tally is a general ledger: it records what you tell it to record, but it does not perform multi-way reconciliation against warehouse physical goods receipts or calculate price-volume variances. Excel sheets lack an immutable audit trail, cannot scale across 1,000 monthly shipments, and introduce human error. Satya connects to Tally and prevents over payments before records are posted."*

Q2: *"Why did you use both MongoDB and Neo4j instead of just PostgreSQL?"*

Answer: *"Procurement data has two distinct shapes. Source records (invoices, challans) are irregular, semi-structured documents best stored

in MongoDB. But the commercial relationships between those documents—multi-hop links, circular advances, and partial credit notes—form an interconnected graph. Neo4j executes multi-hop traversals in constant time, whereas relational SQL joins degrade exponentially as transaction volume grows."*

Q3: *"How can an executive trust your calculations? What if your AI hallucinates a balance?"*

Answer: *"Our AI never performs financial arithmetic. All balance calculations, deductions, and match totals are executed by a deterministic Python engine using 60-digit decimal precision and standard

rounding rules. Machine learning is used only to suggest candidate line matches, which require human author ization."*

Q4: *"Can a tenant ac cidentally see another c ompany's procure ment data?"*

Answer: *"No. Data isolation is enforced at three levels: (1) cryptogra phically via JWT claims, (2) in application middleware where every query is scoped by `c ompanyId`, and (3) in database indexes where `companyId` forms the prefix of all composite keys."*

Q5: *"What happens when supplier d ocuments directly c ontradict each other?"*

Answer: *"Satya does not arbitrarily guess which

document is correct. When a conflict occurs—such as two invoices referencing the same transfer or a PO currency differing from an invoice currency—Satya flags a `PAYMENT_SOURCE_CONFLICT` or `CURRENCY_CONFLICT`, sets the canonical outstanding to indeterminate, and alerts the AP manager to request vendor clarification."*

Q6: *"How do you handle Indian tax compliance and GST regulations?"*

Answer: *"Satya's rules engine integrates standard Indian procurement compliance: 15-digit GSTIN structure validation, HSN directory prefix checks, Form GFR-12A statutory reporting, and MSME 45-day

payment deadline tracking under Section 43B(h)."*

Q7: *"What if an enterprise client refuses to upload financial data to the cloud?"*

Answer: *"Our entire stack is containerized using Docker. Satya can be deployed fully on-premises within a client's private VPC, ensuring that financial documents and records never leave their corporate firewall."*

Q8: *"How do you get data out of legacy desktop accounting software like Tally?"*

Answer: *"In our MVP, we ingest standard CSV and Excel exports that Tally already produces with one click. On our

roadmap, we use Tally's local XML-over-HTTP interface to synchronize vouchers automatically in the background."*

Q9: *"Why would a supplier agree to a deduction calculated by Satya?"*

Answer: *"Because Satya generates an audit-ready executive PDF brief citing the exact warehouse goods receipt number, gate pass timestamp, and photos/sign-offs of the physical inspection. When presented with immutable evidence, disputes are resolved in minutes rather than weeks."*

Q10: *"What prevents Zoho or SAP from building this exact feature to

**morrow?"*
Answer:
"SAP focuses on global Fortune 500 enterprises with budgets exceeding \$100k, while Zoho focuses on broad SME general accounting. Neither specializes in deep, forensic multi-way procurement reconciliation for Indian industrial supply chains. Our defensibility comes from our proprietary reconciliation graph, our calibrated conformal match models, and our workflow integrations."

11.2 Three Codebase Weaknesses Discovered & How to Address Them

1. **Background Job Queue:** Currently uses an in-process worker (`startWorker()`). *Remedy:* Move to BullMQ/Redis before

onboarding more than 3 concurrent enterprise accounts.
2. **Neo4j Fallback Mode:** If Neo4j is offline, graph features degrade to standard tabular views. *Remedy:* Implement persistent graph caching in MongoDB snapshots so UI views remain interactive even during Neo4j restarts.
3. **Automated Tally Connector:** Currently relies on CSV/Excel uploads. *Remedy:* Package a lightweight local desktop bridge agent for seamless Tally synchronization.

12. THE FINAL CTO REVISION CHEAT SHEET

10 Facts You Must Memorize

1. **Reconciliation Precision:** 60-digit decimal precision (`getcontext().prec = 60`) in Python; zero floating-point math.
2. **Snapshot Integrity:** Every snapshot is identified by a SHA-256 hash of its input manifest, rules, and model version.
3. **Core Formula:** $\text{Supported Balance} = \text{Accepted Goods Value} - \text{Linked Payments}$.
4. **Flagship Case:** Meridian Pump Supply (`PO-1001`): 100 ordered, 80 received, 100 billed = ¹18,160 over-billing detected.
5. **Dual Database Roles:** MongoDB for semi-structured document storage; Neo4j for multi-hop graph lineage.

6. **AI Safety Rule:** ML suggests matches; ML *never* moves money or approves invoices automatically (`automaticApproval: False`).
7. **RAG Safeguard:** The Copilot uses a `Fact Registry`; unverified numbers generated by LLMs are rejected.
8. **Communication Outbox:** Multi-channel dispatcher with Indian Quiet Hours (22:00–08:00 IST) and TRAI DLT SMS compliance.
9. **Target Market:** 1,50,000 Indian SME manufacturers and industrial distributors (¹10 Cr to ¹250 Cr turnover).
10. **Unit Economics:** Target Growth Plan ¹6,499/month; Gross margin > 80%; Payback period < 4 months.

5 Technical One-Liners

- *"We use Python Decimal arithmetic with 60 digits of precision to eliminate floating-point drift."*
- *"Every number on screen links directly to a document cell via an immutable SHA-256 snapshot citation."*
- *"Our GraphRAG copilot decouples language from arithmetic: numbers come from tools, words from the LLM."*
- *"Multi-tenant isolation is cryptographically verified at the API gateway and scoped on every MongoDB query."*
- *"Our graph projection models multi-hop circular settlements that relational databases fail to trace efficiently."*

5 Business One-Liners

- *"Satya AI tells a company what it actually owes and proves it with supporting evidence."*
- *"We don't replace your accounting software; we prevent it from paying bad invoices."*
- *"We turn weeks of month-end vendor disputes into an instant, evidence-backed 2-minute decision."*
- *"Mid-market manu facturers lose 1.5% to 3% of procurement spend to billing errors—Satya recovers that cash."*
- *"Our pricing starts at ¹1,999/month with a free historical audit pilot that proves ROI on day one."*

5 Claims You Must NOT Make

- 'L *"Our AI guarantees 100% fraud

prevention."* (Truth: We detect evidence discrepancies; fraud determination requires legal investigation.)

- 'L *"We replace Tally or SAP entirely."* (Truth: We are a specialized reconciliation layer that sits on top of existing ERPs.)
- 'L *"Our ML approves invoices automatically without human review."* (Truth: Corporate policy strictly requires human confirmation.)
- 'L *"We have automated direct bank integrations running in production today."* (Truth: Currently supported via statement ingestion and transactional outbox webhooks.)
- 'L *"We have thousands of enterprise customers."* (Truth: We are a high-conviction early-stage startup with a fully working,

tested MVP.)

End of CTO Master Briefing · Document Persisted as `SATYA_AI_CTO_BRIEFING.md`